Arista Access Switch (HEB-ACC-11) Hardware/Environmental Fault Troubleshooting and Resolution SOP

Node: HEB-ACC-11 | **Vendor:** Arista | **Severity:** Major | **Alarm Code:** ALARM_2278

Description

This Standard Operating Procedure (SOP) is applicable to the Arista Access Switch (HEB-ACC-11) at the Hebbal Edge site, which is experiencing a major failure due to an ASIC memory parity error, line card crash, or power supply redundancy loss. The Arista Access Switch is part of the Access domain, playing a critical role in forwarding traffic from the edge to the core network. The failure of the Access Switch can lead to network instability, packet loss, and service disruptions. The primary reason for this failure domain is the inherent complexity of the Arista ASIC, which is prone to memory parity errors due to its high-performance and high-density design. Additionally, the line card crashes can occur due to hardware or software failures, and the power supply redundancy loss can happen due to a faulty power supply unit or a power grid issue. The root cause of this failure must be identified and resolved promptly to ensure network uptime and reliability. The following SOP outlines the diagnostic and resolution steps to troubleshoot and fix the hardware/environmental fault on the Arista Access Switch.

Pre-requisite Checks

- check: Verify the site location and equipment ID to ensure that the correct device is being worked on., description: Confirm that the Hebbal Edge site and the HEB-ACC-11 device are correctly identified before proceeding with the troubleshooting process.

- check: Check the redundant link status to ensure that there are no network disruptions., description: Verify that the redundant link is up and functioning correctly to avoid any network disruptions during the troubleshooting process.

- check: Confirm that the necessary permissions and access rights are granted to perform the troubleshooting and resolution steps., description: Ensure that the required permissions and access rights are granted to perform the diagnostic and resolution commands on the Arista Access Switch.

Potential Root Causes

Cause	Probability
ASIC memory parity error	High
Line card crash	Medium
Power supply redundancy loss	Low

Diagnostic Commands & Expected Outputs

```
Command: show version, Expected Output: Arista 7250CX-64, Description: This command displays the device model, software version, and hardware information to verify the device configuration and identify potential issues.

Command: show system resource, Expected Output: Memory: 64 GB, CPU utilization: 50%, Description: This command displays the system resource utilization, including memory and CPU utilization, to identify potential memory or CPU-related issues.

Command: show environment, Expected Output: Temperature: 25°C, Humidity: 50%, Description: This command displays the environmental status, including temperature and humidity levels, to identify potential environmental issues affecting the device.
```

Resolution Steps

Step 1: Step: Clear the ASIC memory parity error, Command: clear arithmetic-failure, Description: This command clears the ASIC memory parity error and resets the device to its default configuration. However, this may cause a temporary network disruption.

Step 2: Step: Replace the faulty line card, Command: replace line-card slot 1, Description: This command replaces the faulty line card with a new one and reinitializes the device. This step requires physical access to the device and may cause a network disruption.

Step 3: Step: Verify the power supply redundancy, Command: show power supply, Description: This command displays the power supply status to verify that the redundant power supply is functioning correctly.

Post-Check Verification

```
Command: show version, Expected Output: Arista 7250CX-64, Description: This command verifies that the device model and software version match the expected configuration after the resolution steps.

Command: show system resource, Expected Output: Memory: 64 GB, CPU utilization: 20%, Description: This command verifies that the system resource utilization has returned to normal after the resolution steps.

Command: show environment, Expected Output: Temperature: 25°C, Humidity: 50%, Description: This command verifies that the environmental status has returned to normal after the resolution steps.
```

Roll-back Procedures

- Step: Restore the device to its previous configuration, Command: rollback to savepoint 1, Description: This command restores the device to its previous configuration before the resolution steps. This step requires caution to avoid overwriting any critical configuration changes.

- Step: Replace the original line card, Command: replace line-card slot 1, Description: This command replaces the original line card with the faulty one to restore the device to its original configuration.